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Equitable access to satellite broadband services: Challenges and opportunities for developing countries

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1. Introduction

Universal and meaningful digital connectivity (UMC) is crucial in achieving the Sustainable Development Goals (SDGs) adopted by the United Nations (UN) General Assembly in 2015. Recently, its importance as a public utility was verified when the countries that have achieved digital transformation have been more adept at tackling the challenges brought about by the COVID-19 pandemic than those on the other side of the digital divide (Broadband Commission for Sustainable Development, 2022). This devastating period has further expanded the global development gap and heightened the urgency of addressing the digital divide (Balfour et al., 2022). However, building the underlying broadband Internet infrastructure is a formidable task (Oughton et al., 2023; Schneider-Petsinger, 2022). The recent surge of mega satellite constellations, most of which are deployed in the Low Earth Orbit (LEO), has been met with enthusiasm due to their ability to deliver broadband Internet services in underserved regions and unhospitable terrain by significantly reducing the time and investment required for terrestrial infrastructure. At the same time, the prospect of dependence on infrastructure owned, controlled and operated by non-domestic public or private enterprises for connectivity has raised sovereign autonomy-related concerns (Suess, 2023). As more spacefaring countries compete to deploy their 'sovereign' constellations to mitigate these concerns and reassert their power in outer space and the global Internet infrastructure, the perspectives of developing countries that have much to gain from this technology are largely ignored (Pekkanen, 2019; van Eijk, 2022). The scope of their rights under international law remains contested and inadequately explored.

Ensuring that the connectivity capacity of mega satellite constellations contributes to bridging the global development gap rather than further entrenching it requires developing nations to advocate a coherent policy approach across international law disciplines based on the principles of equity and to engage with the relevant global governance processes and institutions effectively. The two key areas of international law relevant to space-based communication systems are outer space law and telecommunications law. In both disciplines, the principle of equitable access has been linked to the recognition of space as the province of all humankind. The Outer Space Treaty (OST) starts by acknowledging this principle in Article 1, stating that the exploration and use of outer space shall be carried out for the benefit and in the interests of all countries (United Nations [UN], 1967). Building on the principles of this treaty and recognizing that the Earth's orbits are limited natural resources, its commercial and governmental use has been coordinated at the international level, primarily relying on notification and registration requirements. However, mega-constellations pose risks that go beyond these measures. The increase in private ownership and control of space-based communications infrastructures and the space sustainability concerns due to the exponential increase in the number of objects in the orbits, including space debris and the elaborate number of satellites deployed for mega-constellation projects, has raised wide-ranging concerns and revealed the shortcomings of both

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fields of law. These developments also impact developing countries and their present and future potential to benefit from space resources. Norm development is ongoing, most significantly at the International Telecommunications Union (ITU), and it is crucial for developing nations to take part in the processes to safeguard their interests and to advocate an equitable regulatory intervention.

The norms and policies that apply to the mega satellite constellations globally are also relevant to the much-debated right to Internet access and the future-oriented Internet governance agenda (De Hert & Kloza, 2012). The disparity in ownership and control in the commercial space communications industry parallels that of the broader global Internet infrastructure, where the same group of states' public and private actors hold the majority of influence (Ruiz & Barnett, 2015). The Internet Governance processes and the multistakeholder platforms, including technical ones, that have been created to address the problems arising from these disparities should be utilized to promote equitable global policies regarding satellite broadband services, especially concerning their role in the global digital divide and their potential to mitigate or exacerbate related concerns. Consolidated under the Internet Governance Forum (IGF), the recent emphasis on ensuring that 'the benefits of information and communications technologies, including new technologies, are available to all, …and striving to provide universal and affordable access …." (UN General Assembly [UNGA], 2019) is apt to guide the process.

This article aims to contribute to the discussion on the global governance of mega satellite constellations as part of global Internet infrastructure. It argues that (1) equitable access to meaningful and universal connectivity is a legitimate interest of developing countries in participating in norm development processes for satellite broadband, (2) principles of equity in outer space law, telecommunications law and Internet governance policies complement one another and are applicable to the governance of mega satellite constellations, (3) the norm development should defer to agreed principles of equity in international treaties and the sustainable development goals. (4) the coherence of promoted policies across fields will enhance their chance of impact at multilateral and multistakeholder platforms (5) in consideration of the wide range of stakeholders involved and the inherently international nature of space-based connectivity, the problems are best addressed through a global governance perspective, for which international law is a core component.

Section 2 introduces the research questions, methodology, and data. Section 3 establishes the connection between equitable access to the Internet and the SDGs as a legitimizing factor for non-spacefaring nations to participate in norm development for satellite broadband. It also analyzes the significance and the scope of domestic regulations in achieving country-specific UMC goals and needs. Sections 4 and 5 explore the principles of equity that safeguard the rights and interests of developing countries in the context of outer space law and international telecommunications law, respectively. Section 6 discusses the relevance of multistakeholderism in norm development for satellite broadband, with a particular emphasis on the role of the IGF in developing equitable Internet policies. The article concludes in Section 7 with a list of recommendations.

2. Research questions and methodology

This article poses the following questions: 1. What are the ways in which the principle of equitable access can guide the development of a shared understanding and language about the global governance of mega-constellations so that the resulting model for benefit sharing contributes to UMC and sustainable development? 2. What are the most relevant stipulations associated with this principle in the laws of outer space and telecommunications? 3. How can developing countries utilize multilateral platforms and multistakeholder Internet governance processes to influence norm development for the commercial space-based broadband Internet sector, where their presence is currently minimal? In addressing these research questions, the classification of 'developing countries' is not based on strict criteria. Instead, it refers to all countries, primarily non-spacefaring countries, which lack the technological or financial resources, skills, and infrastructure for space exploration and deploying their own satellite systems to improve connectivity within their borders and achieve their SDGs.

This research primarily relies on an analytical analysis of legal and policy documents to interpret and develop the legal practice and to achieve policy objectives. Our extensive desk research involves treaties, domestic regulations, industry policy documents, non-binding international law instruments, current practice and academic literature related to sustainable development goals, outer space law, and telecommunication regulations.

In addition, this research is part of a larger project for which the authors have conducted.

1. Four semi-structured in-depth interviews with academics and practitioners who have deep expertise in policy, development, technical and legal fields: The decision was made not to record the interviews, with the understanding that while the researchers can utilize the opinions and information shared, they must not disclose either the identity or the association of the contributors. Herein, they are referred to as Interviewee(s).
2. Presentation and testing of our findings as shorter publications, ten different workshops and academic conferences: We have received feedback from other experts and non-expert audiences and reevaluated our conclusions in a feedback loop.

The expert interviews and the presentations at various venues have confirmed that our legal argument and policy recommendations are timely and relevant to the problems at hand.

3. Equitable access to the Internet and SDGs

The developing world has a legitimate interest in using space-based connectivity to stay abreast with the rapidly changing digital ecosystem. In recognition of its enabling role for all 17 SDGs, connectivity has sometimes been referred to as SDG Zero (Roesch, 2021).

The Digital Acceleration Agenda recently corroborated this claim, indicating that 70% of 119 SDG targets will benefit from connectivity (ITU, UNDP, 2023). There is also a specific reference to building information and communication technologies (ICT) infrastructure in SDG 9c, which recognizes its transformative potential. However, mere universal availability, however important it may be, is insufficient to ensure equitable access and, consequently, the achievement of the SDGs. Connectivity can fulfil its enabling role when it is available to all and provides '*the possibility for everyone to enjoy a safe, satisfying, enriching, productive, and affordable online experience*' (Broadband Commission for Sustainable Development, 2020).

It is necessary to consider the broader set of universality metrics and connectivity enablers determined by the Broadband Commission, a joint initiative by the ITU and UNESCO, which promotes the idea of broadband as a fundamental global public utility, to understand the mega satellite constellations' impact on SDGs. Accordingly, the key enablers for UMC include the availability and quality of infrastructure, affordability of connection, access to Internet-enabled mobile or fixed devices, digital skills, and the security and safety of the connection (ITU and UN Office of the Secretary General's Envoy on Technology, 2022). For example, if the operator provides services directly to consumers, the users will also need to purchase terminals to connect their devices. Both the price of the devices and the service subscription fees will impact the affordability of satellite broadband services, and affordability can constitute significant binding constraints to meaningful access (Garrity & Garba, 2020). If utilized to complement and expand the reach of existing mobile networks or to provide connectivity collectively to communities, including the price in the existing subscription fees could mitigate the affordability concerns to an extent. These business models are still in their early stages. If they are to enhance global development, their financial viability concerns must be balanced against their affordability. It can also make sense from a commercial perspective. Space X, the pioneering company in the global satellite broadband industry, had initially announced one fixed price to apply globally but then decided to adopt variable pricing depending on local market conditions and parity in purchasing power (Press, 2022). Even if availability, quality and affordability benchmarks are met, other digital skills and safety can constitute significant barriers for a population to benefit from connectivity (Vosloo, 2018). While these factors are equally important, satellite broadband-specific assessments of these benchmarks are beyond the scope of this article.

Domestic regulations serve as a significant tool for countries to implement their country-specific policies, including those related to SDGs, and this is no different for satellite communications (ITU, 2019). They may enhance or impede accessibility, affordability and the time frame in which these systems become available in a country. No uniform domestic regulatory model would fit all, but aligning policy goals regionally and globally will enhance their success. Understanding the implications of the regulations on the policy implementation requires a basic grasp of the technology. The typical architecture of these systems consists of three main components: the constellation of satellites, a network of ground stations, and user terminals (see Fig. 1). In this section, we analyze each component with respect to the legal regime applicable to their use, as well as the radio frequency spectrum enabling the connection between the components. Each analysis reveals domestic control points which could be utilized by domestic regulatory agencies to implement their policies to achieve the key enablers for UMC by adding satellite broadband to their Internet infrastructure.[1]

The Internet services provided by mega satellite constellations are on par with the advancements in broader communication technologies that have continually raised the bar for the desired speed and capacity for UMC, primarily owing to placing a large number of satellites in LEO that operate as integral parts of a unified system. One satellite moves over an area, and another follows a predetermined path to facilitate the continuity of coverage. The previous satellite systems could not meet the speed and latency requirements of real-time web-based applications because they have been primarily deployed in higher altitudes. A single satellite in the Geosynchronous Earth Orbit (GSO) has much larger surface coverage on Earth owing to its altitude of 35 786 km from the Earth's equator, giving it the appearance of being stationary in the sky when viewed from the ground. However, the data has to travel longer distances. As a result, GSO satellite technology has played a marginal role in providing global Internet access (Liu et al., 2021).

In their pre-launch, in-orbit and end-of-life phases, satellite systems are subject to both the Outer Space Law and the domestic laws of the states involved in the launch and international registration of the system. The Outer Space Law provides the basic framework for all international space activities. At the same time, the domestic laws ensure compliance with these and may impose additional requirements, depending on the nature of the satellite system or the service it will provide. The states that are mere beneficiaries of the system do not have control over this part of the operation. Despite this limitation, these countries could still set and implement goals as their licensing and authorization benchmarks, such as using space sustainability metrics for licensing. Establishing a set of principles to guide the behavior of operators could facilitate a more equitable system of access (Deplano, 2023). Presuming that the nations that own and operate it benefit most from the use of space resources, Section IV analysis identifies ensuring its use is available to others under sufficiently good conditions, which should include a commitment to ensure universal availability, reliability and security of the connection at an affordable price on a non-discriminatory basis which will be in line with the Broadband Commission's targets for UMC, as a manifestation of equitable benefit sharing. Embracing shared policy perspectives can potentially foster collaboration with like-minded actors at international platforms where pertinent norms are deliberated and established.

The ground stations act as a gateway to the Internet. LEO satellite companies must comply with domestic laws and regulations to obtain rights to set up ground stations. At the time of this article, the distance between ground stations is required not to exceed about a thousand kilometres to provide continuous coverage. If a country has a smaller surface area and there is a ground station within proximity within the borders of one of their neighbors, technically, they may use that instead of investing in one. However, this means relying on that neighbor not to disrupt your services. There may be additional cybersecurity implications as well (Creedon, 2012). If a country has a large surface area, there may be a need for more than one ground station. Suppose the country wishes to create a

[1] Graphic by: Huihui et al. 2021. © CC BY 4.0.

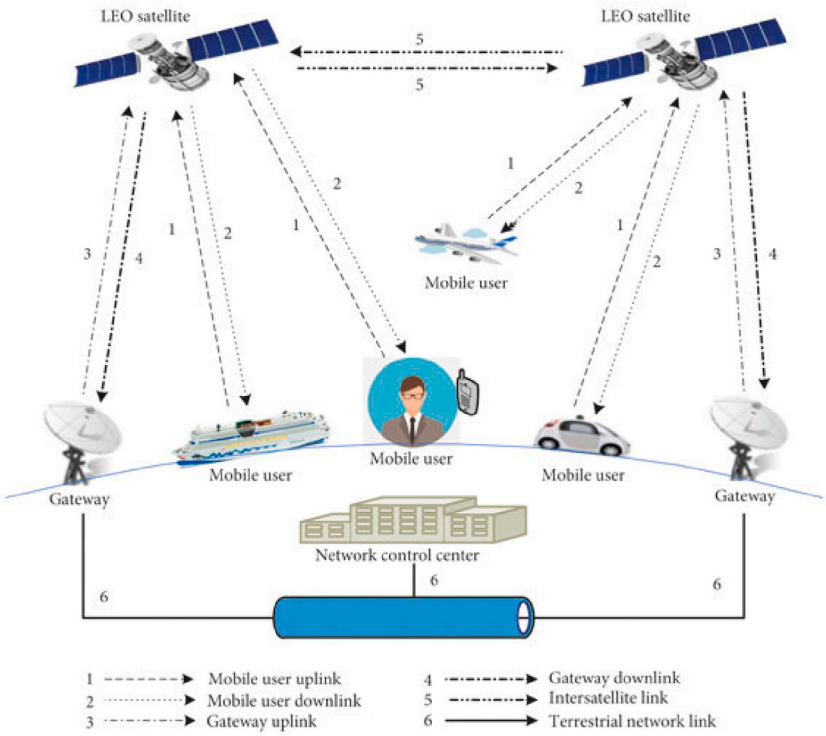

Fig. 1. A low Earth orbit satellite communications system.

competitive environment by authorizing multiple satellite broadband companies. In that case, they will need to arrange the location of these ground stations to avoid interfering with each other's services and other wireless services in the area. The domestic regimes applicable to ground stations have the potential to provide developing nations with the leverage to implement and negotiate a range of UMC targets, especially if conducted through alliances with neighbouring states. Regional and geographical dynamics will be determinant factors for ground station policies. The updates to the regulatory frameworks of spacefaring nations with extensive technological and cyber expertise over these systems can serve as valuable guides. Consultations with various stakeholders are essential to shape these regulations that will have political, technical, security and competition-related implications.

The third component is the user terminal, by which the users connect their devices to satellite broadband services. The satellite company operating the system produce them under the relevant regulations of the authorities where the company is located. However, importation into another jurisdiction will be subject to the importation requirements of the host country, which may include different technical specifications or security requirements, taxes, and duties. Global and regional trade regimes should be an essential aspect of relevant policies. For instance, consumers may be unable to access services even if satellite operators offer them at an affordable price if there are import restrictions on user terminals or unreasonable taxes and duties on other Internet-enabled devices. These requirements can significantly impact device access, affordability and, consequently, UMC. As recently exemplified in the Ukraine-Russia War, when Ukrainian connectivity was disrupted as a result of an attack, the flagship company of this technology, SpaceX, was able to make its services available in days upon demand by the Ukrainian authorities by sending in the user terminals necessary to access its network (Simonite, 2022). In non-emergency situations, the procedural and technical requirements typically take longer. They are more complex. This example demonstrates the key role user terminals play in determining the minimum time required for deployment capability. Time is particularly crucial when addressing global equity concerns, as is the urgency to bridge the digital divide. Therefore, the domestic regulators should ensure that the regime applicable to the user terminals should be coherent with their broader Internet policies.

The domestic regulators typically also require the satellite service provider to obtain authorization to use the radio frequency spectrum to communicate with the Earth, which often requires a reimbursement, sometimes determined through auctions. National regulation of frequency spectrum and the associated licensing procedures within their territories to meet national policy objectives lie within the exclusive authority of each country. However, there is coordination at the international level. The allocation of radio frequency spectrum requires a periodic technical assessment of the changing needs of all wireless communication systems within a country to devise the most efficient way to ensure UMC. In many developing countries, mobile broadband (3G or higher) is often the primary, if not the only, means of Internet connectivity (Clement, 2021). Managing the spectrum requires technically advanced domestic policies, particularly determining the appropriate allocation criteria among various wireless communication technologies such as broadcasting, remote sensing, and military. The broader issues associated with the radio frequency spectrum are further explored in Section V. It is one of the most significant regulatory issues that concerns a wide range of stakeholders. Equitable access and equitable

benefit sharing goals are directly linked to how the radio frequency spectrum is regulated at international and domestic levels.

There may be a range of other regulations imposing additional licensing and authorization requirements, depending on the policy preferences of each jurisdiction. Some of these will impact their trading partners and have world trade law implications. They will be assessed concerning each jurisdiction's commitments under World Trade Organization treaties and preferential trade agreements. As a general policy objective, adopting rational and efficient licensing and authorization requirements and avoiding additional market access requirements will enhance their potential to achieve UMC (Garrity and Husar, 2021).

Domestic regulations have progressively adapted to accommodate technological advancements, global political changes, and the privatization and commercialization of the telecommunications and space sectors (Walden, 2018). Integrating mega satellite constellations to the global communication networks and ensuring equitable access to their services represents a pivotal moment for policy and regulatory evolution, where domestic regulators will assume a crucial role. The analysis in this section defined how the domestic regulatory space could be utilized in a way that aligns with the equitable access and equitable benefit sharing principles analyzed in the next two sections.

4. Principles of equity in outer space law

Outer space law is the body of international legal norms developed primarily under the auspices of the UNGA and its Committee on the Peaceful Uses of Outer Space (COPUOS). The deployment of satellites over the years by an increasing number of spacefaring states and their commercial enterprises has been based on this legal framework, which acknowledges that the orbits are free for use by all states on the basis of equality and in accordance with international law. Considering the significant discrepancy between spacefaring states' technological and economic capacity to invest and engage in space-based activities, the OST also imposes a commitment for these activities to be carried out for the benefit and in the interests of all countries, irrespective of their degree of economic or scientific development (UNGA, 1997). Jakhu argued that OST Article I sets forth a principle of global public interest in outer space (Jakhu, 2006). However, there are no agreed criteria for how benefits accrued from space endeavors should be shared or an understanding of how to conduct a space activity in the interest of all countries. More specifically, there is no specific understanding of what spacefaring countries' owe' to others from their space ventures under this provision (Scanlon, 2000). The growing involvement of commercial actors in space activities, some of which are multinational, complicates matters further. This section aims to address a key aspect of this indeterminacy in the context of mega satellite constellations.

The ideal approach to sharing and preserving our finite global common resources enhances human well-being to the greatest extent. In line with this understanding, the initial space-based telecommunication services were delivered by (hybrid) international organizations and were not owned by commercial interests. In the early 1990s, communication satellites INTELSAT, INMARSAT, and EUTELSAT were non-profit intergovernmental organizations dedicated to providing services on a non-discriminatory (access for all) basis (UNGA, 1961). Although ownership structure had caused concern, these organizations positioned themselves to work in the general public interest (Lyall & Larsen, 2018, pp. 281–335). This structure of the early satellite communication organizations served broader purposes, such as the peaceful use of space, demonstrated by the USSR and the Eastern Bloc joining INTELSAT from the 1980s to the 1990s. After this period, the perception of telecommunications services started to change. As privatization spread, it was perceived less as a public service and more as a service available to the public where profit was a significant objective. One outcome of this change has been that the service availability in an area has been determined by profit, not by the needs of potential users. Telecommunications services provided via satellites shared the same fate, and the intergovernmental organizations were privatized one after the other.

Mega satellite constellations have emerged in an already privatized global commercial market where a wide range of commercial services provided by satellites, owned publicly and privately, benefit a large number of countries. In practice, the accessibility of services generated by satellites by all has been the most common way of benefit sharing; however, the conditions for equitable benefit sharing change in tandem with societal context and time (Franck, 1998). Therefore, the formulation of new policies should take into account the current dynamics of the space sector and the associated concerns about our shared future, essentially achieving SDGs while preserving space sustainability. While the possibility of the reemergence of non-commercial systems was a frequently raised question at conference presentations of this research, both the authors and the experts interviewed are skeptical about the prospect. Nevertheless, the pursuit of fairness can continue to guide and safeguard the non-commercial interests of stakeholders if appropriately defined and grounded in established norms and advocated and implemented through long-standing multilateral mechanisms.

Over the years, the UN has reiterated its intention to support the equitable utilization of orbital resources and universal access to space-based communication services by making satellite communications available where they are needed, as soon as practicable on a global and non-discriminatory basis (UNGA, 1961; UNGA, 2001). These aspirations were also detailed in the Vienna Declaration on Space and Human Development in 1999 (UN, 2019UN, 2019). Accordingly, a democratic and equitable international order will facilitate the use of space technology to benefit all nations and peoples and extend its applications to developing countries (UNGA, 2003). Most recently, the UN's Space 2030 Agenda corroborates this understanding. In an endeavor to facilitate the use of space technologies for development, the documents set forth action items that seek to balance commercial and noncommercial interests, including the interests of future generations (UNGA, 2021). However, reaching a consensus to produce binding instruments to address the aforementioned contemporary space governance problems has been difficult (Johnson, 2020). UNCOPUOS, however, has been able to produce a set of non-binding guidelines and resolutions that shed light on how OST principles should be interpreted in the context of space sustainability and debris mitigation (Hertzfeld et al., 2016). These are the most relevant stipulations of UN action confirming the applicability of the equitable access principle with respect to satellite connectivity.

At this stage, the commitment to equitable access to orbital resources can be realized partially by benefit sharing from using these

resources, primarily equitable access to broadband services. As proposed in Section III, assurance of universal, reliable availability of these services to others at a reasonable cost and without discrimination will be a step towards equitable access. Referring to UMC enablers to promote equity will set a common understanding when discussing the role of satellite broadband in global communications, particularly in relation to the principle of equitable benefit-sharing under OST.

The space sustainability problems, particularly those related to mega-satellite constellations such as space debris and space traffic, pose potential barriers to equitable access to orbital resources and the benefits derived from their use (Cappella, 2019; Virgili, 2016). Civil society and diplomatic efforts are underway within multilateral and multistakeholder platforms (Organisation for Economic Co-operation and Development [OECD], 2022). The space debris problem exacerbated by the rapid increase in satellite constellations, both currently in orbit and planned for launch, was a recurring concern among our interviewees and conference attendees. These concerns are well-founded, given that at the time of this research, various countries had collectively filed for over a million satellites across more than 300 constellations. With only about 8000 active satellites currently in orbit, which already pose a management challenge, these concerns are justified (Falle et al., 2023). From a connectivity standpoint, it is crucial to adopt a sustainable approach to orbit management and gather more evidence regarding the capacity of the existing systems and the number of constellations needed to achieve universal connectivity (Allnutt & Pratt, 2019). The interpretations and application of principles of equity should defer to the significance of preserving orbital sustainability in consideration of the severity of the challenges created by mega satellite constellations.

5. Equitable access principle in international telecommunications law

In recognition of the linkage between connectivity and development, the ITU's development sector ITU-D was created in 1991, and its mandate to provide technical assistance to developing countries was placed on the same level as its role in standardization and spectrum management. The ITU's different roles are complementary. Its regulatory role in spectrum management and orbital resources is to be carried out with respect to its mandate to achieve equitable communications for everyone, reach universal connectivity, and bridge the digital divide. Therefore, the ITU's responsibility to facilitate equitable access to ICTs and broadband satellite connectivity does not need to be inferred from the right to equitable access to space resources under OST or equitable benefit sharing to achieve fairness. It has been mandated expressly to this institution as part of its mission (ITU Constitution Article 44(2) & ITU Radio Regulations [RR] No 0.3). In keeping with its mandate, the ITU assumes an active role in global development and addressing emerging policy issues arising from the changing telecommunication environment, including satellite broadband. The ITU has been at the center of discussions regarding the promises and perils of mega-constellations, especially because of the operational role of its radiocommunications sector (ITU-R), responsible for the coordination of radio frequency spectrum and orbital space, both of which are limited natural resources.

The ITU-R has a system governing the use of GSO, guaranteeing developing countries equitable access once they have reached technological and economic development to become spacefaring nations themselves (Altholz, 1986; Way, 2018). Its regulations regarding satellites in lower orbits, referred to as non-GSO orbital space, were less burdensome, and an a priori planning was not deemed necessary. As constellations with large numbers of satellites started to cause concern, the ITU-R changed its regulations at the World Radio Conference (WRC) in 2019. In Resolution 35 (WRC-19), the term constellation was defined for the first time as "*non-GSO satellite systems having more than one orbital plane where the mutual relative position of each orbital plane and each satellite in its orbital plane is important*." An indicator of whether the system represents a constellation was added as a mandatory data item to the filing system under the same resolution (Resolution 35, Appendix 4). The requirement for the coordination procedure has been redefined to include constellation projects. A new time-phased approach regarding the time frame allowed for the deployment of constellations has also been introduced, which is designed to help ensure that its Master International Frequency Register is aligned with the actual deployment of non-GSO satellite systems. Through these regulatory interventions, the ITU aimed to "*establish the balance between the prevention of spectrum warehousing, the proper functioning of coordination, notification and registration mechanisms, and the operational requirements related to the deployment of non-GSO systems*" (Resolution 35 (f)). This resolution marked the first substantial revision to the ITU regulations, in accordance with the principle of equitable access and aimed essentially to achieve a fairer and more equitable allocation of the orbital resources and the associated frequencies. However, it achieved only limited success. It unintentionally promoted spectrum warehousing, an unfair practice where radio frequencies and associated orbital resources are reserved without being actively used, thereby blocking others from utilizing these resources. Stricter requirements to reserve and maintain allocated radio frequencies and associated orbital resources will likely discourage such practices (Falle et al., 2023). If not mitigated, this type of exploitation of the ITU system will prevent the use of limited global resources in a way that maximizes its positive societal impact (ITU, 2022).

The regulations were renegotiated and updated at the time of writing at WRC 2023, where a new resolution titled 'Sustainable Use of Radio-Frequency Spectrum and Associated Satellite-Orbit Resources' was endorsed. This document acknowledges and supports the critical role of the ITU in space sustainability, including providing guidance on safe and efficient strategies and methodologies for the deorbit and disposal of non-GSO space stations, once again in line with the equitable access principle, observing the needs of the developing countries. Furthermore, the document encourages members to collaborate and share information with other UN organizations involved in space activities to avoid duplicating efforts. The regulatory manifestation of this document is anticipated to set tolerance limits on certain orbital characteristics of non-GSO systems and include a milestone procedure in orbital space by consistently decreasing the number of satellites in orbit after milestones have been achieved (Zoller, 2023). These two resolutions have been the most recent stipulations of its ITU actions concerning the fair use of the orbital space and radio frequency spectrum, which has an impact on not only satellite broadband but all wireless communication technologies. They show that the ITU, particularly ITU-R, is

critical to satellite broadband governance primarily because it updates its regulations and adapts to technological changes every 3–4 years at the WRCs, with the help of contributions from multistakeholder working groups. Its technical and operational role will remain critical to enduring equitable access to satellite broadband, associated orbital resources and radio-frequency spectrum.

Determination of the terms of fair access to the services provided by satellite broadband companies is a separate matter that falls more squarely under the ITU-D's official mandate to "*promote the availability of infrastructure and foster an enabling environment for telecommunication/ICT infrastructure development and its use in a safe and secure manner*." A comprehensive strategy, grounded in principles of equity as suggested in this study, will be the key to linking the allocation of resources to satellite broadband technology and fairness in its global use.

6. The role of multistakeholder Internet Governance

Satellite broadband is also in the purview of Internet governance actors and systems. Their goal is to, in their respective roles, develop and apply shared principles, norms, rules, decision-making procedures, and programs that shape the evolution and use of the Internet, whether they are governments, the private sector or civil society (Working Group on Internet Governance, 2005). The principle of multistakeholderism is integral to the governance of the Internet. The UN set up the Internet Governance Forum (IGF) in 2006 to facilitate policy dialogue on Internet governance issues among the stakeholders. This model aims to counter the power imbalance in ownership and control of the global digital realm by a small number of states and their private companies by offering all stakeholders an equal role in protocol development and policymaking. The effectiveness of the IGF and the multistakeholder model has been controversial. Its transparency and inclusiveness have been questioned (Kleiman, 2020; Mueller & Badiei, 2019). While the model has fueled Internet growth, its limitations have become apparent (Kulesza, 2018). Nonetheless, it has become essential in global governance mechanisms (Weber, 2020). Mega constellations mark the most recent frontier for the multistakeholder approach to Internet governance. The introduction of sustainable development-oriented regulation will require effective functioning of the multistakeholder governance processes amidst the swift development of this sector and their resistance against the pull towards top-down regulation.

The goal of addressing problems associated with satellite broadband connectivity falls squarely within the original IGF mandates in the Tunis Agenda, particularly its mandate in paragraph 72(e), which is to 'advise all stakeholders in proposing ways and means to accelerate the availability and affordability of the Internet in the developing world'(ITU, 2005). As of 2015, the IGF also endorsed SDGs and committed to undertaking its technical and public policy mandate in conformity with them. This step complements the Tunis Agenda promise for equitable, human rights-based universal access to the Internet and its infrastructure. It also acknowledges the vital role ICTs play in national development, often in conjunction with digital strategies. There are significant challenges to undertaking this task. Mega-constellations have not only amplified the problems related to space-based communication infrastructures but also the broader political and technical challenges tied to global Internet governance, including associated public and private power dynamics.

Space technologies' control, use and ownership have always been associated with strategic, economic, technological and military power. Satellites are considered geo-political infrastructures, whether used for communication or other services. The new generation mega satellite constellations are rapidly becoming integral to global broadband Internet networks. Their much-publicized use in the Ukraine-Russia war to facilitate Internet access to Ukrainians to counter restrictive actions by Russia also confirmed their geo-political significance (Akcali Gur & Kulesza, 2022). The prevalence and power of private actors constitute a further complication. These preexisting and new concerns regarding space-based infrastructures contribute to the increased securitization of global communication infrastructures and Internet governance. Internet governance issues, as well, have long been important for geo-political and geo-economic strategies (DeNardis, 2020). Control and influence over the Internet's technical infrastructure have been recognized as proxies for controlling social, economic, and political spheres (DeNardis & Musiani, 2016). The expansion of connectivity transformation confirmed the significance of the physical infrastructure, including telecommunication networks, as one of the most critical assets. The ownership and control of telecommunications technologies have emerged as critical issues related to sovereignty, security, and power (Dunn Cavelty & Wenger, 2020). Satellite broadband has become a part of this complex interplay of geographical and political factors that define connectivity geopolitics.

The necessity to strike a balance between competing interests, including national security, global cybersecurity, commercial goals, and UMC for achieving SDGs, while urgently implementing strategies to maintain space sustainability not only highlights the multidimensional and multistakeholder nature of the problem but also underscores its delicate political and security dimensions. In this context, the pivotal role of states in developing norms will continue to be significant, which aligns with the current Internet governance conjuncture. In the decade following the World Summit on the Information Society in 2005, the discussions on specifying the notion of Internet governance, most significantly through defining the respective roles of states, businesses, and civil society, intensified. Especially for this debate, Musiani, Cogburn, DeNardis, and Levinson have distinguished between technical and political Internet governance, with a rapid "turn to infrastructure" enabling states to increasingly restrict the scope of policymaking left to the multistakeholder community through regulation (Levinson et al., 2016; DeNardis & Musiani, 2016). Contemporary academic discourse offers a few novel solutions to this challenge (Radu & Chenou, 2014; Sylvan, 2014). The legal, technical and policy experts whom we interviewed were also not opposed to bringing the satellite broadband debate to global multistakeholder fora. However, they expressed doubts about these platforms' capacity to yield practical outcomes. The feedback to our research presentations has revealed an enthusiasm to learn and discuss the relevant issues in the context of multistakeholder engagement.

Given the significant presence of private interests in the satellite broadband sector, exemplified by companies like SpaceX, where a single shareholder holds a significant stake or by multinational ownership structures like OneWeb, the role and influence of private actors should also be considered in the global governance of satellite broadband. They could either pose a challenge or serve as a

beneficial counterbalance in achieving equitable access to their services. At this nascent stage of satellite broadband technology, where prominent examples in the sector are mostly based on one leading company, SpaceX and its Starlink satellite constellation, the drawbacks of privatized Internet governance are perceived to outweigh the benefits (Kradomski, 2022). The prospect of critical communication infrastructures being subject to the decisions of a single individual is understandably viewed as a risk. These companies are operating globally and extending their services across jurisdictions through partnerships with other private companies across the connectivity ecosystem and governments that provide connectivity as a public service. Considering that this is a critical infrastructure, foreign dependence is already perceived as a challenge, even if its control is not left to one person. However, these contractual relationships could also be perceived more positively in the context of Bygrave's take on private Internet policymaking. Bygrave pointed out that the largest group of the global population is governed more by the power of private contracts than by multilateral agreements. He proposed that these contractual arrangements could provide solutions to numerous Internet governance issues by acting as a balancing factor to its increased securitization and state-centric turn (Bygrave, 2015). While it is still early to determine the impact of the prevalence of private actors in the global governance of satellite broadband, the established extensive role of states and multilateral organizations, the need for global coordination of space resources, and the specific security-related status of satellites suggest that the influence of private interests will likely be gradually decreased and will be balanced against competing interests.

Multistakeholder participation in policymaking is associated with procedural fairness. However, it is insufficient in and of itself to eliminate the underlying geo-economic and geo-political frameworks that influence decision-making in conventional domestic and multilateral configurations and to establish an even playing field for all parties, resulting in a just distribution of benefits and burdens. These circumstances are rooted in global power dynamics among the actors, which change depending on the economic, social, and political environment (Raymond & DeNardis, 2015, p. 572). Yet, the relationships in various multilateral and multistakeholder engagement models can impact the resulting policies and norms if engaged effectively in advocating an SDG-oriented policy strategy (Coolsaet & Pitseys, 2015). Hence, nations that stand to benefit from satellite broadband technology should not undermine the multistakeholder platforms. Instead, they should seize the engagement opportunities therein to advocate their policy approach to protect their interests. Using existing global processes and the international law framework about how their policies are determined and their rules are made, interpreted, and applied will ensure their legitimacy.

7. Conclusion and recommendations

The role of satellite broadband in the global communications industry has not yet matured. It is on the path to becoming a more integral part of the broader global communications network, complementing the existing cable and wireless infrastructure. Developing countries, particularly those impacted most by the global digital divide, must ensure fair terms of access to this service to achieve UMC. So, they have a legitimate interest in participating in the relevant global norm development processes. This endeavor requires robust and informed policies that will shape their regulatory interventions. In contributing to this timely debate, this research proposes that the equitable access principle in international telecommunications law and parallel iterations in outer space law be re-assessed and applied to justify the terms of fair access and, more broadly, the developing country rights to benefit from satellite broadband services. Grounding the overarching policy framework in this principle, one that is recognized in the aforementioned international law fields, will ensure its legitimacy when advocated in relevant multilateral and multistakeholder platforms.

The analysis in this research demonstrated the connections between the developing countries' right to demand fair terms of access to satellite broadband with the right to share benefits generated by the utilization of orbital space - a limited common resource and the preservation of space sustainability. The orbital sustainability concerns are exacerbated as a result of the increasing number of mega satellite constellation infrastructures. Developing country policies aimed at achieving UMC should not produce results that compromise space sustainability, which is an equally important concern that impacts their present and future potential to benefit from space resources. The ongoing initiatives of the ITU-R to ensure equitable access to the radio frequency spectrum and orbital space allocation, examined in Section V, exemplify the intertwined concerns and competing interests of developing countries in relation to sustainable development, connectivity, and the use of outer space resources. Balancing overlapping and competing interests requires policies in different fields to be formulated and pursued in conjunction with each other. For advocating these developing country interests, relying on principles of equity is best suited for achieving coherence.

The policy framework proposed herein is more likely to succeed if it's endorsed by developing countries themselves within multilateral platforms and multistakeholder Internet governance processes, where norms for satellite broadband are negotiated and developed. Currently, the participation of developing countries in these processes is limited, and their influence is minimal. The comprehensive analysis conducted in the respective sections of this article has revealed potential action items, which are presented in the following paragraphs.

First, developing countries should define their distinct needs and interests concerning their domestic, regional, and global connectivity ecosystem. They should assess these and the corresponding domestic policies and regulations and renew them where necessary to maximize the contribution that newly enhanced satellite connectivity can make. In this respect, their power to determine the domestic rules and regulations governing the licensing and authorization of frequency spectrum and ground stations and other aspects of satellite broadband services analyzed in Section III will be the foremost tool to integrate them into their existing infrastructure in a manner that will enhance their connectivity.

Second, national regulatory agencies, primarily those governing issues related to telecommunications, development, and space agencies, should align their policies to ensure a consistent approach to the domestic connectivity ecosystem, including satellite broadband. These goals are more likely to be achieved if agreed upon at regional and global levels and adhered to and served

previously agreed upon principles, such as the principle of equitable access to achieve UMC and the SDGs. A shared understanding and language becomes significant, especially in the context of ITU-R negotiations, one multilateral organisation that has been actively pursuing regulatory responses to address satellite broadband-related challenges. Informed and effective participation is necessary for all countries, especially developing countries primarily relying on wireless communication technologies. Collaboration within regional groups or groups that share similar concerns can be instrumental in pooling resources and expertise to represent shared interests effectively. The recent regulatory updates analyzed in Section V reveal the ITU's commitment to the application of principles of equity in the context of the use of orbital space and radio frequency spectrum by mega satellite constellations. The authors propose that these regulatory interventions and proposals for revisions should not be thought of in isolation from the goals and mandate of the ITU-D and the global SDGs.

Fourth, developing countries should mitigate their lack of expertise in space technologies to define relevant interests and goals by effectively utilizing UNOOSA's resources to build capacity, promote sustainable development, and ensure the sustainability of outer space activities. The governance of outer space is a matter of concern for all countries, even those non-spacefaring. They are current co-beneficiaries of commercially available space-based services, including satellite broadband, and they are also potential future spacefaring countries. Informed decisions to join outer space treaties can grant nations additional rights, and non-spacefaring countries should prioritize this option. The collaborative efforts through regional agencies could potentially enhance their chance of advocating for the fair distribution of benefits derived from the utilization and exploitation of space resources by others. The authors proposed in Section IV, relying on their analysis of previous UN resolutions, that demand for UMC enablers, universal availability, reliability and security of the connection at an affordable price on a non-discriminatory basis is justifiable based on equitable benefit-sharing requirements.

Fifth, the authors concluded that developing countries should participate in multistakeholder processes. They should use or pool their resources to take advantage of the multistakeholder Internet governance platforms, primarily IGF, to promote domestic and regional satellite broadband-related issues, focusing on affordability, reliability, and security as central factors of equitable access. Prior practice, as analyzed in Section VI, revealed the significance of their contribution to the ongoing discussion and engagement with other stakeholders to address joint challenges. Their aligned efforts to establish best practices based on the equitable access principle will enhance the coherence and transparency of space-based broadband policies across relevant sectors.

Future studies that delve into the initiatives of civil society and the effects of regulatory measures taken by leading spacefaring countries will enhance this research and guide developing countries in their efforts to ensure equitable access to space resources, radio frequency, and global Internet infrastructure.

CRediT authorship contribution statement

Berna Akcali Gur: Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Joanna Kulesza:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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References

Akcali Gur, B., & Kulesza, J. (2022). *Satellite internet access in times of cyber Conflict*. Commentary. *Directions* Retrieved from: https://directionsblog.eu/satellite-internet-access-in-times-of-cyber-conflict/. (Accessed 22 November 2023).

Allnutt, J. E., & Pratt, T. (2019). *Satellite communications*. John Wiley & Sons.

Altholz, E. D. (1986). Warc 1985: The effects of an equitable access regime on satellite telecommunications services. *1 University of Chicago Legal Forum, 10*.

Balfour, R., Bomassi, L., & Martinelli, M. (2022). *Coronavirus and the Widening global North-South gap*. Carnegie Endowment for International Peace.

Broadband Commission for Sustainable Development. (2020). *Manifesto: Global goal of universal connectivity*.

Broadband Commission for Sustainable Development. (2022). *The State of Broadband: Accelerating broadband for new realities*.

Bygrave, L. A. (2015). *Internet governance by contract*. Oxford University Press.

Cappella, M. (2019). The principle of equitable access in the age of mega-constellations. In M. Cappella (Ed.), *Legal aspects Around satellite constellations* (pp. 11–23). Germany: Springer.

Clement, J. (2021). Mobile internet traffic as percentage of total web traffic 2021, *Statista*. Retrieved from https://www.statista.com/aboutus/our-research-commitment/408/j-clement. (Accessed 10 October 2022).

Coolsaet, B., & Pitseys, J. (2015). Fair and equitable negotiations? African influence and the international access and benefit-sharing regime. *Global Environmental Politics, 15*, 38–56.

Creedon, M. R. (2012). Space and cyber: Shared challenges, shared opportunities. *Strategic Studies Quarterly, Spring*, 3–8, 2012.

De Hert, P., & Kloza, D. (2012). Internet (access) as a new fundamental right. Inflating the current rights framework? *European Journal of Law and Technology, 3*.

DeNardis, L. (2020). Internet governance as an object of research Inquiry. In L. DeNardis, et al. (Eds.), *Researching internet governance* (pp. 1–20). Ma, USA: MIT Press.

DeNardis, L., & Musiani, F. (2016). Governance by infrastructure. In F. Musiani, et al. (Eds.), *The turn to infrastructure in Internet governance* (pp. 3–21). New York: Palgrave MacMillan.

Deplano, R. (2023). Inclusive space law: The Concept of benefit sharing in the outer space treaty. *International and Comparative Law Quarterly, 72*(3), 671–714.

Dunn Cavelty, M., & Wenger, M. (2020). Cyber security meets security politics: Complex technology, fragmented politics, and networked science. *Contemporary Security Policy, 41*, 5.

Falle, A., Wright, E., Boley, A., & Byers, M. (2023). One million (paper) satellites. *Science, 382*, 150–152.

Franck, T. M. (1998). Fairness and international law: An analytical framework. In T. M. Franck (Ed.), *Fairness in international law and institutions*. Oxford, UK: Oxford Academic.
Garrity, J., & Garba, A. (2020). *The Last-mile internet connectivity solutions guide: Sustainable connectivity Options for unconnected Sites*. International Telecommunication Union.
Garrity, J., & Husar, A. (2021). *Digital connectivity and low Earth orbit satellite constellations opportunities for Asia and the Pacific*, 76 ADB Working Paper Series, 1-30.
Hertzfeld, H. R., Weeden, B., & Johnson, C. D. (2016). Outer space: Ungoverned or lacking effective governance? New Approaches to managing human activities in space. *The SAIS Review of International Affairs, 36*(2), 15–28.
Huihui, H., Xuyang, M., Zehui, W., & Qiang, W. (2021). An efficient ECC-based authentication Scheme against Clock asynchronous for spatial information network. *Math. Prob. Eng.*, 1–14.
ITU. (2005). *Tunis agenda for the information society*. Retrieved from http://www.itu.int/wsis/docs2/tunis/off/6rev1.html. (Accessed 9 October 2022).
ITU. (2019). *Global internet growth stalls and focus shifts to 'meaningful universal connectivity' to drive global development*. Press release, 22 September 2019.
ITU. (2022). *Regulation of satellite systems Retrieved from*. https://www.itu.int/en/mediacentre/backgrounders/Pages/Regulation-of-Satellite-Systems. (Accessed 9 October 2022).
ITU and UN Office of the Secretary General's Envoy on Technology. (2022). *Achieving universal and meaningful digital connectivity: Setting a baseline and targets for 2030*. Retrieved from at www.itu.int/itu-d/meetings/statistics/wp-content/uploads/sites/8/2022/04/UniversalMeaningfulDigitalConnectivityTargets2030_BackgroundPaper.pdf. (Accessed 9 October 2022).
ITU, UNDP. (2023). *SDG digital acceleration agenda*. Retrieved from https://www.sdg-digital.org/accelerationagenda (Accessed 11 November 2023).
Jakhu, R. S. (2006). Legal issues relating to the global public interest in outer space. *Journal of Space Law, 32*, 31–110.
Johnson, K. (2020). *Key governance Issues in space. Report*. Center for Strategic and International Studies (CSIS).
Kleiman, K. (2020). Crash goes ICANN's multistakeholder model. *American University Intellectual Property Brief, 11*(1), 2–18.
Kradomski, A. (2022). *Democratic infrastructure must not be owned by the few*. Chatham House. Expert Comment https://www.chathamhouse.org/2022/11/democratic-infrastructure-must-not-be-owned-few. (Accessed 9 October 2023).
Kulesza, J. (2018). Balancing privacy and security in a multistakeholder environment. ICANN, WHOIS and GDPR. *The Visio Journal, 3*, 49–58.
Levinson, L., N.S, & Cogburn, D. (2016). The next "turn" in Internet infrastructure governance. In F. Musiani, et al. (Eds.), *The turn to infrastructure in Internet governance*. New York: Palgrave MacMillan.
Liu, S., Gao, Z., Wu, Y., Wing Kwan Ng, D., Gao, X., & Wong, K.-K. (2021). LEO satellite constellations for 5G and beyond: How will it Reshape Vertical Domains? *IEEE Communications Magazine, 59*, 30–36.
Lyall, F., & Larsen, P. B. (2018). *Space law A Treatise*. Oyfordshire, UK: Routledge.
Mueller, M. L., & Badiei, F. (2019). Requiem for a dream: On advancing human rights via internet architecture. *Policy & Internet, 11*, 61–83.
OECD. (2022). *Earth's orbits at risk: The economics of space sustainability*. OECD Publishing.
Oughton, E. J., Amaglobeli, D., & Moszoro, M. (2023). *What would it cost to connect the unconnected? Estimating global universal broadband infrastructure investment*. Telecommunications Policy, Article 102670.
Pekkanen, S. M. (2019). *Governing the new space Race. 113 AJIL Unbound 92*.
Press, L. (2022). *SpaceX introduces affordability-based Starlink pricing. CircleID.com* Retrieved from: https://circleid.com/posts/20220906-spacex-introduces-affordability-based-starlink-pricing. (Accessed 10 November 2023).
Radu, R., & Chenou, J. M. (2014). From Nested Dilemmas to democratic internet governance. In R. Radu, J. M. Chenou, & R. H. Weber (Eds.), *The evolution of global internet governance: Principles and policies in the making* (pp. 191–198). Germany: Springer.
Raymond, M., & DeNardis, L. (2015). *Multistakeholderism: Anatomy of an Inchoate global institution. 7 international theory*. ff.
Roesch, J. L. (2021). *Sdg Zero? A People-Centered approach to universal connectivity*. International Peace Institute. Report. New York.
Ruiz, J. B., & Barnett, G. A. (2015). Who owns the international Internet networks? *Journal of International Communication, 21*(1), 38–57.
Scanlon, T. M. (2000). *What We owe to each other*. Harvard University Press.
Schneider-Petsinger, M. (2022). Closing the digital infrastructure and connectivity gap. In L. Vinjamuri, B. Lee, M. Malloch-Brown, J. O'Neill, H. Clark, C. L. Liao, … M. Marianne Schneider-Petsinger (Eds.), *Building global prosperity: Proposals for sustainable growth* (pp. 64–72). London: Royal Institute of International Affairs.
Simonite, T. (2022). How Starlink Scrambled to Keep Ukraine Online. *Wired*. Retrieved from: https://www.wired.com/story/starlink-ukraine-internet/ Accessed on 23 November 2023.
Suess, J. (2023). *Starlink 2.0? Taiwan's plan for a sovereign satellite communications system*. RUSI Commentary. Retrieved from https://rusi.org/explore-our-research/publications/commentary/starlink-20-taiwans-plan-sovereign-satellite-communications-system. (Accessed 9 November 2023).
Sylvan, D. (2014). Global internet governance: Governance without governors. In R. Radu, J. M. Chenou, & R. H. Weber (Eds.), *The evolution of global internet governance: Principles and policies in the making* (pp. 23–37). Springer.
UNGA. (1961). Resolution 1721 (XVI). In *International Cooperation in the peaceful uses of outer space, 55/122*. Part D: A/RES/.
UNGA. (1997). *Declaration on international cooperation in the exploration and Use of outer space for the benefit and in the interest of all states, taking into particular account the needs of developing countries*. A/RES/51/122.
UNGA. (2001). *International cooperation in the peaceful uses of outer space*. A/RES/55/122.
UNGA. (2003). *Promotion of a democratic and equitable international order*. A/RES/57/213.
UNGA. (2019). *Information and communications technologies for sustainable development Report of the Second Committee*. A/74/378.
UNGA. (2021). *The "Space2030' agenda: Space as a driver of sustainable development*. A/RES/76/3.
United Nations. (2019). *The space Millennium: Vienna Declaration on space and human development*. A/CONF.184/6-UN, chap. I, resolution 1.
van Eijk, C. (2022). Unstealing the sky: Third world equity in the orbital commons. *Air and Space Law, 47*, 1.
Virgili, B. B., et al. (2016). Risk to space sustainability from large constellations of satellites. *Acta Astronautica, 126*, 154–162.
Vosloo, S. (2018). *Designing inclusive digital solutions and developing digital skills: Guidelines*. UNESCO Digital Library. ED/PLS/YLS/2018/03. Retrieved December 18, 2020, from https://unesdoc.unesco.org/ark:/48223/pf0000265537. (Accessed 9 November 2023).
Walden, I. (2018). International regulatory law. In I. Walden (Ed.), *Telecommunications law and regulation* (pp. 791–845). New York: Oxford University Press.
Way, T. A. (2018). The space gap, access to technology, and the Perpetuation of poverty. *International Research Scape Journal, 5/7*.
Weber, R. H. (2020). A legal lens into internet governance. In L. DeNardis, D. L. Cogburn, N. S. Levinson, & F. Musiani (Eds.), *Researching Internet Governance* (pp. 105–121). London, UK: MIT Press.
Working Group on Internet Governance. (2005). *Report of the working group on internet governance*, 05.41622, 4.
Zoller, J. (2023). Orbital characteristics and operational flexibility in non-GSO space stations. *ITU News Magazine*. Retrieved from https://www.itu.int/hub/2023/10/orbital-characteristics-and-operational-flexibility-in-non-gso-space-stations/. (Accessed 22 November 2022).
United Nations. (1967). *Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space Including the Moon and other Celestial Bodies*, 610 UNTS 205 Article II.